

Statistical-mechanical characterization of memory and quench response in decentralized LLM-agent networks

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Abstract

We study a finite interacting system whose microscopic variables are realized choices made by independent large-language-model agents. Agent i owns a categorical belief $b_i \in \{-1, +1\}$, a committed action $a_i \in \{-1, +1\}$, confidence, commitment, bounded private memory, a local field, workload, inbox, outbox, and typed authority. A pinned instruction model generates random-sequential updates using only local state and delivered messages. We connect these microdynamics to collective order, fluctuations, entropy rate, multi-information, an effective symmetric-layer energy, relaxation, and path-reversal divergence. Earlier discovery and prospective replication studies found memory effects of 0.01790 and 0.04030 nats per attempted update, respectively. We then execute a prospectively frozen field-quench and restoration study comprising 17,280 decisions in 24 trajectories over 6 new matched graph clusters. Field reversal changes maximum post-quench macrostate departure relative to nominal evolution by 133.805 (95% interval 107.949 to 184.075), while restoration changes early-to-late departure by 134.110 (106.654 to 184.196). The full statistical-mechanical representation changes cluster-held-out balanced accuracy relative to order-only variables by 0.333 (0.167 to 0.458), although the simple-uncertainty representation attains the same mean balanced accuracy as the full representation. All six field-reversal clusters recover below their held-out nominal thresholds six sweeps after restoration; the frozen peak-minus-late H3 sign statistic is structurally nonnegative and is interpreted only with the full recovery path. These finite-size, single-model results show what is and is not retained by a reduced statistical-mechanical description. Reference energy is an effective observable, decoding noise is not physical temperature, and path-reversal divergence is coarse-grained temporal asymmetry rather than exact thermodynamic entropy production.

Keywords: statistical mechanics of agents; kinetic Ising models; large language models; entropy rate; irreversibility; quench dynamics.

1 Introduction

Networks of autonomous language-model agents are emerging interacting many-body systems. Each component uses a complicated language-conditioned stochastic rule, retains private state, receives only selected messages, and acts with bounded local authority. The microscopic rule is not a spin-flip formula, yet the network produces familiar collective phenomena: alignment, fragmentation, fluctuations, persistence, and recovery after changes in fields or links. A central question for statistical physics is therefore not whether a

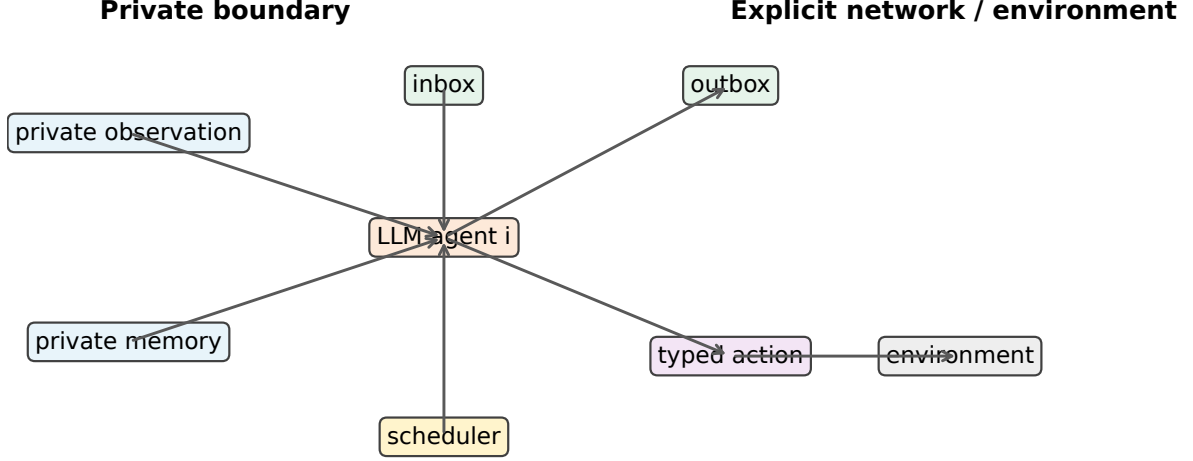


Figure 1: Independent-agent information and authority boundary. The scheduler offers an update and transports the model-selected packet; it cannot inspect peer-private state or substitute a belief, action, commitment, memory, signal, or tool. The reduced state Z_t is evaluator-only.

thermodynamic metaphor can be attached after simulation, but whether a compact set of well-defined observables links local response to collective trajectories and makes distinct regimes quantitatively interpretable.

The kinetic Ising model demonstrates how stochastic local updates generate order and relaxation [1]. Logit dynamics has a related equilibrium interpretation when strategic interactions derive from a potential [2]. Away from detailed balance, state-space currents and path reversal distinguish stationary irreversibility [3–5]. These exact constructions provide useful references, but an LLM network violates several assumptions: the visible binary state may not close under the dynamics; natural-language messages can carry hidden semantic variables; memory induces history dependence; and decoding temperature controls token sampling rather than a physical bath.

Recent studies have nevertheless found compact collective structure in LLM populations. Network topology and social weighting alter conformity [6]; intrinsic field bias can dominate effective cooperation [7]; noise and conformity produce finite-size debate crossovers [8]; population size changes basins of attraction [9]; and large collections of communities exhibit a small number of dynamical regimes predicted by a fitted statistical model [10]. The present study is complementary. It preserves explicit private-state and message boundaries, separates belief from executed action, measures trajectory-level information and fluctuation observables, separates memory discovery from prospective replication, and applies prospectively frozen quenches and counter-quenches to new interacting-agent clusters.

Our objective is characterization, not performance optimization. We ask how bounded memory changes temporal asymmetry; how a field quench and restoration trace paths through a common reduced state; whether topology and channel perturbations follow the same path; and which information is missed by order or uncertainty summaries alone. No observable is required to improve task reward, and no human or field-system inference is made.

2 Statistical mechanics of agent collectives

Statistical mechanics is useful here at three distinct levels. First, realized binary choices define a discrete configuration on a graph. Second, empirical response functions can be approximated by kinetic models and analyzed by mean-field or stability methods. Third, order, fluctuations, entropy, energy, correlations, and time asymmetry reduce a high-dimensional trajectory to a small set of interpretable coordinates. Only the first two ingredients are needed for a rigorous finite stochastic process; no assertion that an LLM is a physical thermodynamic object follows.

Information quantities retain their literal mathematical meanings. Shannon entropy measures uncertainty of a specified distribution [11]. Total correlation measures dependence not captured by marginal entropies [12]. Transfer entropy measures directed predictive information [13]; it is not entropy production. For a stationary Markov chain with observed transition kernel, probability-current entropy production has a precise definition. When visible variables omit history, forward/reverse path divergence is instead a coarse-grained lower-bound diagnostic [14–16]. Memory and hidden slow variables can change the thermodynamic bookkeeping of a reduced process [17], while finite-state time-asymmetry remains constrained by cycle affinities in models where those affinities are defined [18]. Finite-sample entropy estimators are biased, particularly under sparse support, so all such quantities require fixed coarse-graining and null floors [19].

2.1 Relation to kinetic, nonreciprocal, and quench models

The closest analytical reference is an asymmetric kinetic Ising system, for which mean-field closures can be organized around response functions and delayed correlations [20]. Our measured transition rule differs in two ways. Its coefficients are induced by a language-conditioned model rather than specified as a spin kernel, and belief and action form distinct but coupled layers. We therefore test a fitted kinetic closure against direct trajectories instead of assuming it by construction.

Nonreciprocal couplings can generate phenomena absent from equilibrium Hamiltonian systems [21], and their entropy production can be defined for suitable stochastic dynamics [22]. The immutable V12 study used this theory as a reference but found no reliable incremental irreversibility from degree- and traffic-matched directionality in the actual LLM trajectories. V13 preserved that boundary rather than searching new directed graphs, and V14 does not reopen it. The nonequilibrium perturbations are controlled quenches in fields, active delivery support, and packet content. These are finite classical stochastic trajectories; quantum-quench results [23] supply terminology about relaxation but not a claimed mapping of microscopic dynamics.

3 Independent LLM-agent microdynamics

3.1 Observable state and autonomy

The local state is

$$x_i(t) = (b_i(t), a_i(t), c_i(t), q_i(t), m_i(t), w_i(t)), \quad (1)$$

where $b_i, a_i \in \{-1, +1\}$ are categorical belief and action, c_i is bounded confidence, q_i a categorical commitment, m_i bounded memory, and w_i local workload. Each persistent

identity separately owns these variables, a private field h_i , an inbox, outbox, role, and typed authority. The global record $X_t = (x_1, \dots, x_N)$ is available only to the evaluator.

At each attempted update a random permutation schedule offers one agent a turn. Its prompt is rebuilt from its private observation, local state, explicitly delivered packets, and bounded memory when enabled. The model chooses belief, action, confidence, commitment, memory state, outgoing categorical signal, and tool. One syntax-repair attempt is permitted; a persistently invalid response leaves the state unchanged rather than invoking a centralized scientific choice. One sweep equals N attempted updates.

The model is Qwen2.5-7B-Instruct at the pinned revision reported in the data package [24]. The primary regime carries no hidden chat history: the prompt is fully reconstructed from recorded current state. A secondary regime adds exactly three local memory summaries. Neither construction proves Markov closure, which is tested empirically.

3.2 Latent-label counterbalancing

Displayed alternatives “amber” and “cobalt” are semantically symmetric. Their order and mapping to latent ± 1 spins are counterbalanced, private fields and disordered starts are balanced, and all responses are decoded to the latent coordinate before analysis. Matched factor and quench arms share graph, initial state, update schedule, display mapping, and inference seeds. Thus a coupling or disruption contrast does not change option identity or message opportunity.

4 Individual response law

The microscopic grid varies private field h_i , prior belief and action, delivered neighbor field, coupling J , and decoding-noise setting τ . The primary measured responses are realized categorical choices; reported confidence is secondary and is not interpreted as a calibrated probability. We fit

$$\Pr[b_i(t+1) = +1 \mid X_t] = \text{logit}^{-1}(\beta_0 + \beta_h h_i + \beta_n J \bar{b}_{\partial i} + \beta_b b_i + \beta_a a_i), \quad (2)$$

$$\Pr[a_i(t+1) = +1 \mid X_t] = \text{logit}^{-1}(\gamma_0 + \gamma_b b_i(t+1) + \gamma_a a_i(t)). \quad (3)$$

The model is descriptive. It supplies susceptibilities and a surrogate, not a normative Bayesian agent or a claim that token temperature equals an effective thermodynamic temperature.

5 Effective kinetic and mean-field models

For a homogeneous approximation, let m_b and m_a be layer means. Replacing the local neighbor field by m_b yields

$$m_b(t+1) \simeq 2f_b(h + Jm_b(t), m_b(t), m_a(t)) - 1, \quad (4)$$

$$m_a(t+1) \simeq 2f_a(m_b(t+1), m_a(t)) - 1. \quad (5)$$

Linearization of this two-dimensional map gives a local stability matrix. The belief component contains the fitted slope multiplied by J , while persistence and belief–action coupling modify both eigenvalues. We report the local slope as a crossover diagnostic, not a critical exponent.

Writing the fitted logits as

$$u_b = \beta_0 + \beta_h h + \beta_n J m_b + \beta_b m_b + \beta_a m_a, \quad (6)$$

$$u_a = \gamma_0 + \gamma_b \tilde{m}_b + \gamma_a m_a, \quad (7)$$

and $s(u) = 2(1 + e^{-u})^{-1} - 1$, the homogeneous map is $\tilde{m}_b = s(u_b)$ and $\tilde{m}_a = s(u_a)$. Since $s'(u) = 2p(u)[1 - p(u)]$, its local Jacobian is

$$\mathbf{J} = \begin{pmatrix} s'(u_b)(\beta_n J + \beta_b) & s'(u_b)\beta_a \\ s'(u_a)\gamma_b s'(u_b)(\beta_n J + \beta_b) & s'(u_a)[\gamma_b s'(u_b)\beta_a + \gamma_a] \end{pmatrix}. \quad (8)$$

The spectral radius $\rho(\mathbf{J})$ is a local persistence diagnostic. A value near one predicts slow response of the fitted homogeneous surrogate; it does not prove singular behavior in either the finite surrogate or the LLM system.

The CPU surrogate applies the fitted stochastic response to independent agents on ring and fixed-degree modular graphs for $N = 8, 16, 32, 64$. It densely maps coupling and decision-noise settings with 16 graph/seed realizations per cell. Direct LLM observations remain the empirical anchors: the surrogate can explain trend directions or miss them, but cannot replace those trajectories.

6 Macroscopic observables

Belief and action order are

$$m_b = N^{-1} \sum_i b_i, \quad m_a = N^{-1} \sum_i a_i, \quad (9)$$

and cross-layer overlap is $q_{ba} = N^{-1} \sum_i b_i a_i$. Disagreement is the fraction of undirected support edges joining unlike spins. Finite-size belief susceptibility is $\chi_b = N \text{Var}(m_b)$, without a temperature normalization. Integrated autocorrelation time is calculated from the positive sequence of the magnetization autocorrelation.

The fixed symmetric-layer reference energy is

$$H_{\text{ref}} = -\frac{J_b}{2} \mathbf{b}^T W_s \mathbf{b} - \frac{J_a}{2} \mathbf{a}^T D_s \mathbf{a} - K \mathbf{a}^T \mathbf{b} - \mathbf{h}^T \mathbf{b} - \mathbf{g}^T \mathbf{a}, \quad (10)$$

with the reported coordinate $e_{\text{ref}} = H_{\text{ref}}/N$. The communication partition changes active delivery but not this reference layer. We call $N \text{Var}(e_{\text{ref}})$ an energy-fluctuation observable, not physical heat capacity.

For large networks, a fixed macrostate binning of $(m_b, m_a, q_{ba}, e_{\text{ref}}, D)$ defines configuration entropy and a first-history conditional entropy rate. Local-word entropy uses each node's belief, action, and neighbor majority. Total correlation is

$$\mathcal{T} = \sum_i H(X_i) - H(X_1, \dots, X_N), \quad (11)$$

estimated over a five-sweep rolling window. These quantities answer different questions: marginal disorder may be high while dependence is also high, and an entropy increase may represent adaptation, fragmentation, or transient uncertainty depending on order and energy.

The evaluator’s reduced coordinate is

$$Z_t = (m_b, m_a, q_{ba}, S_{\text{config}}, h_{\text{rate}}, \mathcal{T}, e_{\text{ref}}, v_E, \chi_b, \rho, D). \quad (12)$$

It is a prospectively fixed observable representation, not a complete thermodynamic state.

The direct trajectories make this distinction concrete. During the field-reversal interval, mean absolute belief magnetization remains only 0.0042, close to the nominal value 0.0097, because the counterbalanced private fields cancel at the population mean. Nevertheless, rolling configuration entropy rises from 0.223 to 0.387 nats, entropy rate from 0.070 to 0.356 nats per sweep, total correlation from 0.274 to 4.909 nats, and the energy-fluctuation observable from 0.010 to 1.162. A count or mean-spin representation therefore misses a reorganized but globally balanced collective configuration.

7 Prospective design and inference

V12 remains the immutable discovery study. It found that coupling and decoding noise changed order, susceptibility, and persistence, that bounded memory increased adjusted block irreversibility by 0.01790 nats per update, and that traffic- and degree-matched nonreciprocity did not reliably add irreversibility. V13 independently tested those directions. Its coupling and noise contrasts did not replicate, whereas its memory contrast was 0.04030 nats per update in the discovery direction. Its four-cluster quench analysis also showed a large but metric-dependent field-reversal departure and only preliminary representation separation.

V14 does not rerun the broad factor grid. Before any new formal outcome, we froze six independent $N = 16$ modular graph/environment clusters. Within each cluster, nominal evolution, field reversal, removal of inter-community delivery, and corruption of exactly half of eligible categorical packets share the graph, private fields, initial state, update schedule, display mapping, and inference seeds. Every trajectory contains 15 baseline, 15 perturbation, and 15 restoration sweeps. The field-reversal contrast is primary; partition and corruption are secondary comparisons.

The inferential unit is the matched graph/environment cluster. Updates, agents, messages, rolling windows, and tokens are not independent replicates. H2 tests field-reversal maximum departure minus nominal. H3 tests early counter-quench departure minus the final-five-sweep mean. H4 tests the full-minus-order-only leave-one-cluster-out balanced-accuracy contrast. Each uses exhaustive one-sided cluster sign flips, 10,000 cluster-bootstrap interval replicates, and Holm correction across H2–H4. H1 is the already completed V12 discovery–V13 replication memory contrast and is not relabelled as a new V14 experiment. The H3 peak-minus-late-mean contrast is structurally nonnegative except for a constant recovery path. We therefore retain its frozen numerical analysis but do not treat its sign-flip probability alone as evidence of relaxation; threshold crossing, final residual displacement, and the complete recovery trajectory carry the scientific interpretation.

8 Entropy and effective-energy representation

Order and energy are correlated but not interchangeable. Configurations with similar magnetization can differ in domain arrangement, action alignment, multi-information, and temporal predictability. Likewise, a shift in reference energy can arise from field mismatch,

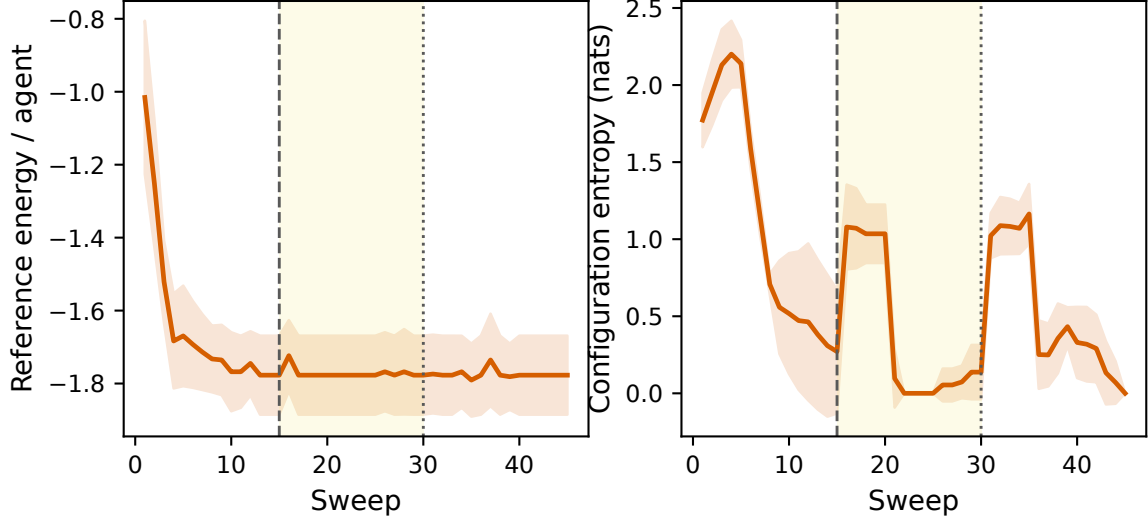


Figure 2: Effective reference energy per agent and configuration entropy through the field quench and counter-quench. Lines summarize complete graph trajectories; the energy is a compatibility coordinate, not literal physical energy.

action–belief inconsistency, or spatial disagreement. Joint energy–entropy trajectories therefore expose degrees of freedom hidden by state counts alone.

We omit a scalar free-energy coordinate. The microscopic response provides a decision-noise control and fitted slopes, but no unique, reproducible physical temperature mapping. The two-dimensional energy–entropy plane is therefore more transparent and avoids implying a Lyapunov function.

9 Persistent memory and irreversibility

For the frozen macrostate word Y_t , the length- L reversal statistic is

$$\mathcal{I}_L = \frac{1}{L-1} D_{\text{KL}}[p(Y_t, \dots, Y_{t+L-1}) \| p(Y_{t+L-1}, \dots, Y_t)]. \quad (13)$$

We subtract the mean from 200 time-shuffled paths of identical length and state occupancy. The result is reported in nats per attempted update. We also measure history-order conditional mutual information. Nonzero history dependence means that $\mathcal{I}_L$ remains a coarse-grained irreversibility diagnostic, not an exact total entropy-production rate.

The immutable V12 discovery effect was 0.01790 nats/update and the V13 prospective replication effect was 0.04030 nats/update. They are reported separately and combined only in a labelled descriptive inverse-variance synthesis. Field, graph, schedule, mapping, initial state, and inference seeds were shared within each memory pair. Fixed block lengths two, three, and four, shuffled bias floors, entropy rates, autocorrelation, and conditional mutual information diagnose whether the effect reflects persistent history rather than state occupancy alone. No new V14 memory trajectory enters either confidence interval.

Numerically, the discovery estimate was 0.01790 nats/update (95% interval 0.00357–0.03399, 24 matched clusters) and the prospective replication was 0.04030 (0.02883–0.05856, six matched clusters). Their explicitly descriptive fixed-effect synthesis is 0.02936 (0.01872–0.03999). This synthesis summarizes consistency; it does not transform the discovery sample into confirmatory evidence or add a new V14 memory experiment.

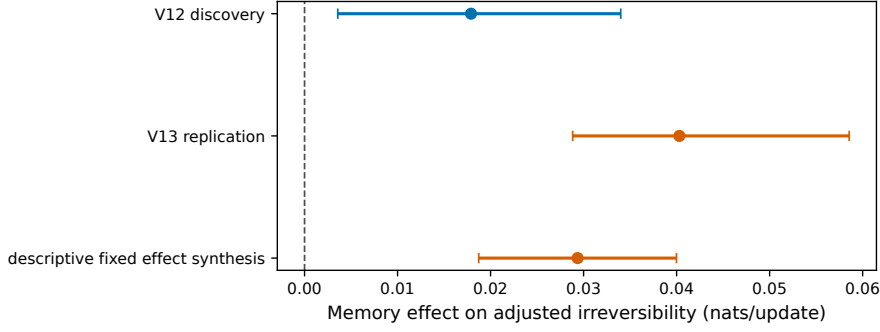


Figure 3: Discovery, prospective replication, and descriptive synthesis of the matched memory effect. Intervals are study-specific; the synthesis does not retroactively pool the two protocols.

10 Controlled quenches and recovery

The three disruptions isolate environmental, topological, and informational changes. Field reversal changes every private field and then restores it. Partition removes cross-community delivery paths while preserving within-group communication and message opportunities. Corruption flips categorical content for a preassigned half of senders while preserving payload length and delivery count. Agents see only the resulting local evidence; the disruption label is not prompted.

A leave-one-cluster-out nominal manifold is fit from undisturbed trajectories using training-cluster-only standardization and a shrinkage Mahalanobis metric on Z_t . For abrupt quenches we measure response—not early warning: maximum departure, path length, time outside the nominal region, overshoot, recovery time, final residual, and the signed route area. Field reversal changes maximum post-quench departure relative to nominal evolution by 133.805 (95% interval 107.949 to 184.075). Its early counter-quench peak exceeds the final-five-sweep mean by 134.110 (106.654 to 184.196). These values are coordinates in the frozen regularized metric, not universal physical scales.

Every field-reversal cluster has a positive H2 paired contrast. Mean maximum post-quench departure is 137.73 for field reversal and 3.92 for nominal evolution (exact one-sided sign-flip $p = 0.015625$; Holm-adjusted $p = 0.046875$). Restoration generates a second departure pulse before all six trajectories return below their held-out nominal thresholds exactly six sweeps later; the mean final-five-sweep distance is 1.56. The frozen H3 contrast is 134.11 (nominal exact $p = 0.015625$; Holm-adjusted $p = 0.046875$), but because peak-minus-final distance is nonnegative by construction, that sign-flip probability is not standalone evidence. The threshold crossing, small residual, and full time-resolved counter-quench path establish the reported relaxation. Partition and corruption remain near nominal in this design, so the result is specific to a private-field quench rather than an omnibus disruption detector.

11 Quantifying information in the reduced state

Three representations are frozen before formal execution. The order-only model uses belief and action magnetization plus belief–action overlap. The simple uncertainty model uses confidence, individual uncertainty, and disagreement. The full model adds configuration entropy, entropy rate, total correlation, mutual information, effective energy, energy

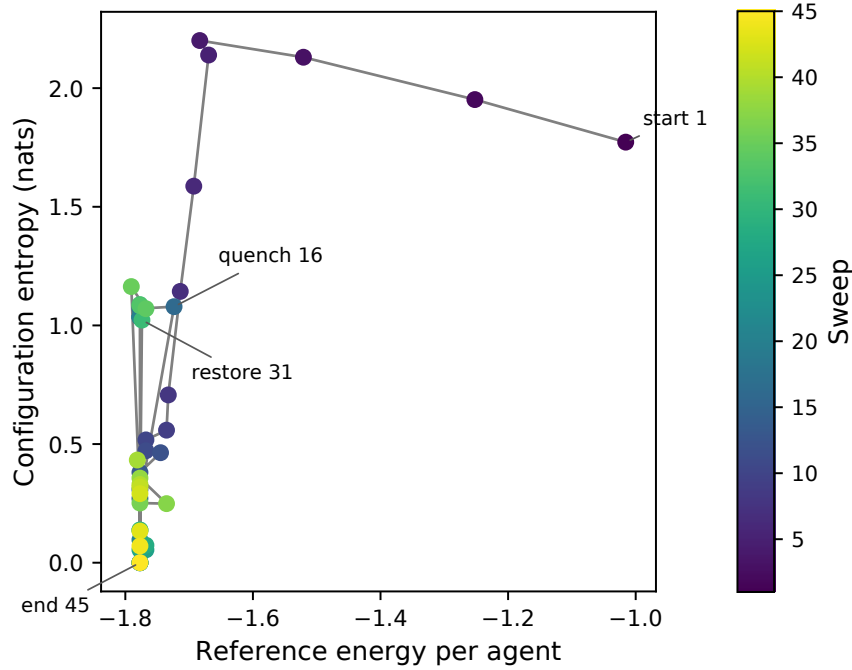


Figure 4: Energy–entropy phase-space path under field reversal and restoration. The color scale and boundary annotations retain temporal direction and distinguish the quench from the counter-quench.

fluctuations, susceptibility, spatial correlation, and temporal coordinates. A regularized multinomial logistic model is trained under leave-one-graph-cluster-out cross-validation; no observable is selected on a held-out trajectory.

The full representation’s cluster-held-out balanced accuracy is 0.458, versus 0.125 for order only; the simple-uncertainty representation also reaches 0.458. The full-versus-order paired difference is 0.333 (95% interval 0.167 to 0.458; exact one-sided sign-flip $p = 0.03125$; Holm-adjusted $p = 0.046875$). Cluster predictions, confusion matrices, feature-family deletions, coefficient stability, and cluster-preserving permutations determine whether any increment is robust or driven by one family. This is not a claim that entropy outperforms a controller; it asks whether jointly measured fluctuation, dependence, compatibility, and temporal coordinates retain trajectory structure absent from order averages.

The result is specific rather than omnibus. The full model identifies all six held-out field reversals, compared with one for order only and five for simple uncertainty, but nominal, partition, and corruption remain mutually confusable. Across 276 prespecified distance sensitivities, 266 preserve a positive field-minus-nominal contrast and ten are exactly zero; none reverse the ordering. Removing the entropy/dependence family nearly eliminates the mean contrast (0.056 distance units), whereas removing energy, order, response, uncertainty, or temporal families preserves a large contrast in the primary shrinkage geometry. This identifies dependence and entropy as important to the distance construction, while the accuracy tie prevents a claim that the full state dominates the much smaller uncertainty representation. The frozen audit did not implement every single-observable deletion or alternative rolling window, and those missing analyses are not reconstructed post hoc.

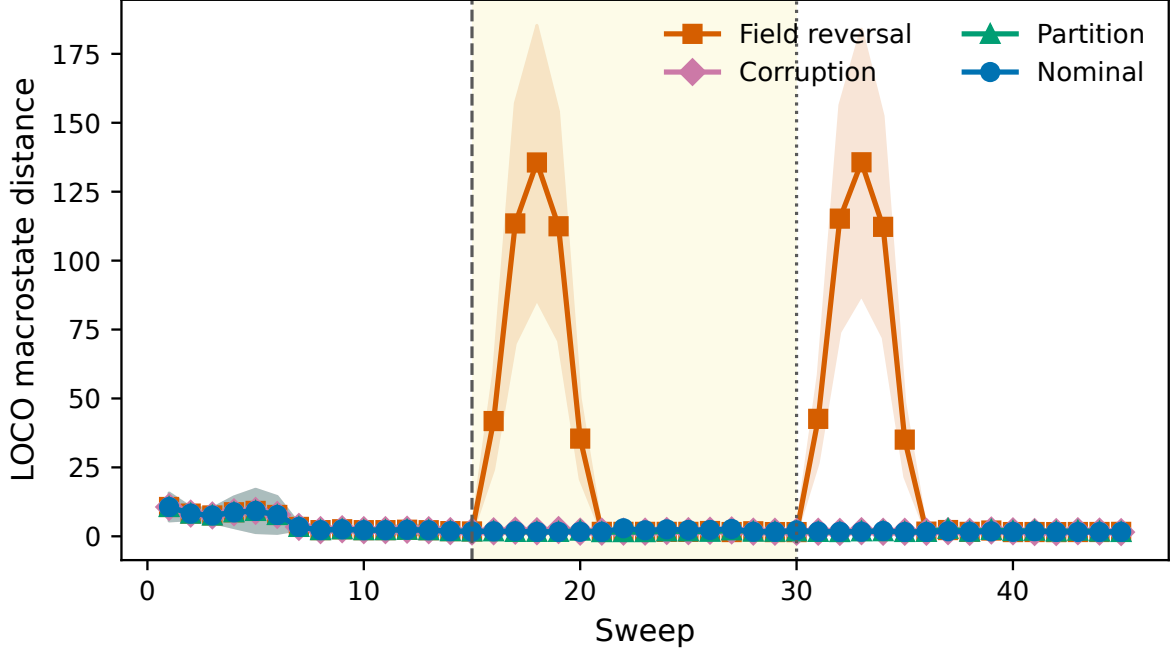


Figure 5: Leave-one-cluster-out distance from nominal operation across four matched conditions. The colored span marks the frozen perturbation interval; uncertainty is across independent graph/environment clusters.

12 Effective-model comparison

The immutable V13 comparison found that the fitted kinetic surrogate missed the direct LLM coupling direction and overestimated the noise contrast. V14 retains that negative boundary. Adding persistence or semantic indicator terms is considered only as a low-dimensional out-of-sample diagnostic, not as a post-hoc fit target. Persistent memory, heterogeneous fields, belief-action lag, saturation, state-dependent response, asymmetric message use, and prompt context are therefore candidate coordinates where the simple closure fails.

13 Limits of thermodynamic interpretation

Several distinctions are essential. First, binary coordinates are measured choices, but the LLM’s hidden computation is not a binary spin. Second, H_{ref} is an effective observable defined by the analyst; no Gibbs stationary distribution is asserted. Third, decoding temperature alters token sampling and is called decision noise. Fourth, total correlation and entropy rate are information measures whose estimates depend on finite windows and coarse-graining. Fifth, time-reversal divergence becomes exact Markov entropy production only under closure and stationarity assumptions not generally met by these data. Finally, $N = 8, 16$ direct networks cannot establish a thermodynamic limit.

The fixed-degree modular topology has a perfect cross-community matching. This is analytically convenient and makes partition restoration exact, but is not a generic social graph. One open-weight model, symmetric abstract alternatives, and short bounded outputs limit external validity. Natural-language semantics are deliberately constrained to obtain a tractable microstate; richer tasks would introduce additional slow and hidden coordinates.

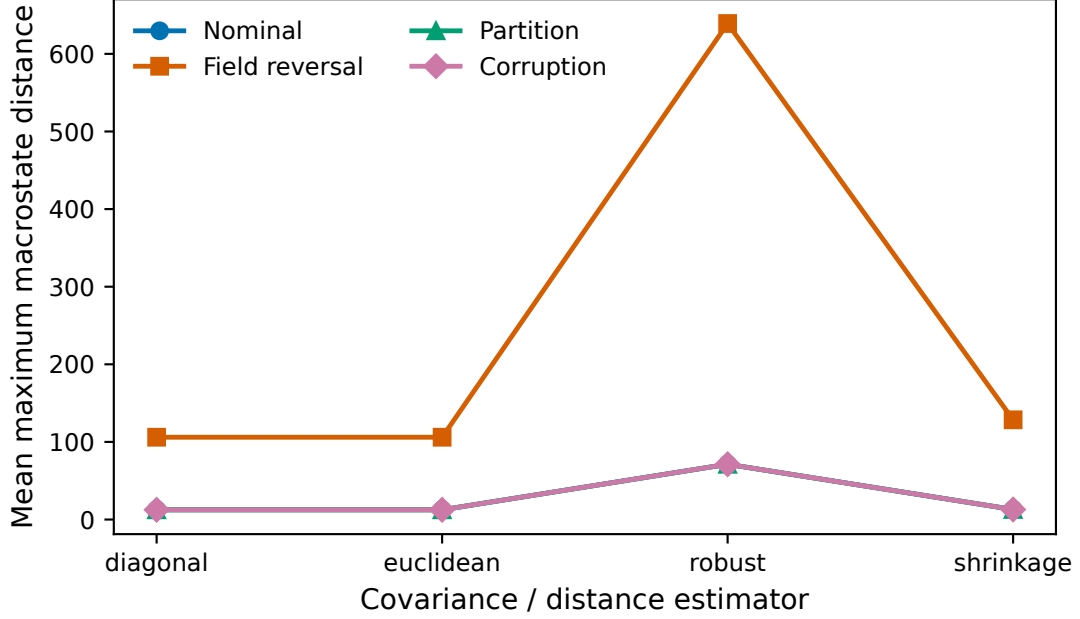


Figure 6: Sensitivity of condition ordering to shrinkage, diagonal, robust, and standardized Euclidean distances. Ridge, nominal-fitting-window, and observable-family-deletion results are retained in the associated aggregate tables.

14 Discussion

The scientific value of this framework lies in compression with explicit boundaries. A local response coefficient connects delivered state to a network coupling. Magnetization identifies order, susceptibility identifies collective fluctuation, autocorrelation identifies persistence, and multi-information separates dependence from local uncertainty. Effective energy combines spatial, cross-layer, and field mismatch. Path reversal detects time asymmetry that may be invisible to static state frequencies. In the balanced field quench, magnetization alone barely moves while entropy rate, total correlation, energy variance, and nominal-manifold distance form two large pulses. The first marks reorganization and the second marks the counter-quench, followed by rapid recovery. The overlap of partition, corruption, and nominal trajectories shows the complementary boundary: a fixed reduced representation does not guarantee sensitivity to a weak or dynamically ineffective perturbation.

The immutable V12 nonreciprocity boundary is also informative. Directed delivery alone was not the dominant detectable source of time asymmetry under matched degree and traffic. In contrast, bounded memory explicitly introduces a historical coordinate and is a plausible mechanism for persistence and irreversibility. This motivates future theory in which message semantics and memory are slow variables rather than treating all asymmetry as an antisymmetric pair coupling.

Future work should add independent models, richer finite-state decisions, larger direct sizes, longer stationary records, and preregistered semantic perturbations. Exact small-state kernels could clarify which projected currents are robust to hidden variables. None of these extensions is needed to interpret the present finite-size measurements, but they are needed before universality or thermodynamic-limit claims.

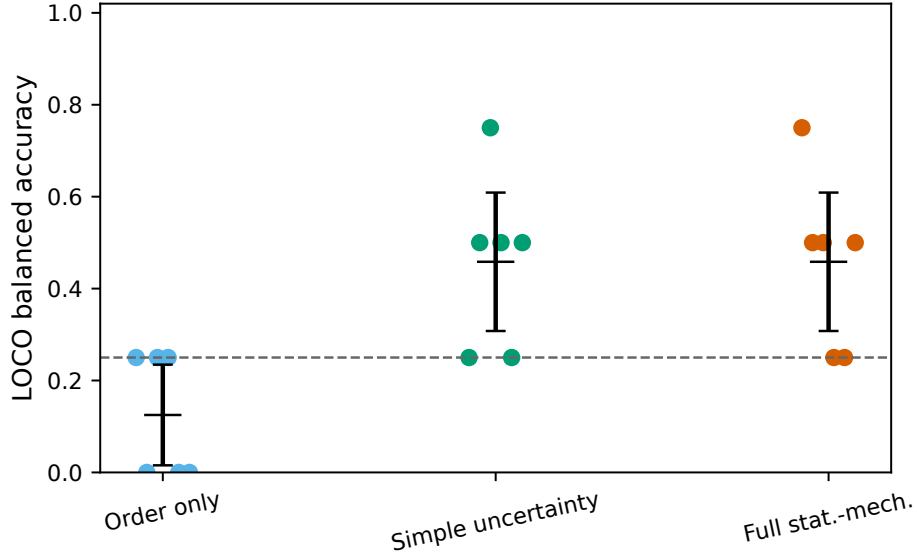


Figure 7: Leave-one-graph-cluster-out disruption classification for simple, order-only, and full statistical-mechanical representations. Points are held-out clusters, and the dotted line is four-class chance.

15 Conclusion

Independent LLM agents with private state and explicit message delivery form a measurable stochastic many-agent process. The V12 coupling and decoding-noise directions did not replicate: V13 estimates were null or reversed, and the fitted kinetic surrogate captured only the noise direction. Bounded memory did replicate, increasing adjusted pathwise irreversibility by 0.04030 nats per attempted update. The reduced energy–entropy–order representation sharply tracked field reversal and improved four-class accuracy over order-only variables, but tied the simple-uncertainty representation and left partition, corruption, and nominal operation weakly separated. Its clearest added value is mechanistic: it resolves dependence and temporal reorganization when the population mean remains close to zero. Statistical mechanics is therefore useful here as a finite-size, reduced descriptive framework with demonstrated memory and quench sensitivity, not as a universal response law or a promise that thermodynamic quantities improve agent performance.

Reproducibility and data availability

The protocol, aggregate tables, figure source data, vector figures, source, tests, checksums, and exact commands accompany the repository. Raw prompts, completions, and attempted-update records are excluded from Git and retained in a checksummed external artifact tree. Formal execution used 7.90 metered generation GPU-hours including the retained pilot. Deterministic replay feeds each content-addressed model decision through the same transition code without another model call. The repository reports model revision, quantization, package versions, tokens, calls, byte counts, exclusions, and every independent graph cluster.

AI-assisted preparation declaration. An AI-assisted coding and writing system was used to help implement the simulation and analysis software, execute the prespecified workflow, check artifacts, and prepare this working manuscript. The submitting authors

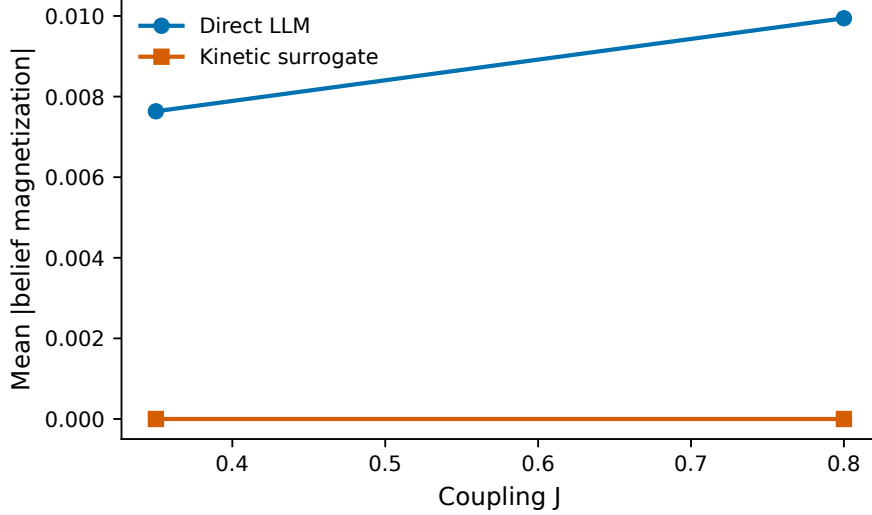


Figure 8: Direct LLM order against the fitted decentralized kinetic surrogate at matched anchors. The diagonal is equality, not a fit target.

remain responsible for the study design, verification, interpretation, citations, and final text. No generated chain-of-thought is retained or treated as evidence.

A Random-sequential transition protocol

For every sweep, a deterministic seed generates a random permutation of all N identities. At opportunity k , only the scheduled agent’s prompt is constructed. Its inbox contains the latest packets actually delivered to that identity; a peer’s private field, memory, and undelivered outbox are absent. The validated model object is decoded through the panel’s latent-label map and applied to that agent. Its selected packet is serialized, one recipient is drawn from the current delivery row, and delivery occurs before opportunity $k + 1$. Consequently, updates within one sweep are temporally linked, while each identity receives exactly one opportunity per sweep.

Matched arms reuse graph, initial state, private fields, label map, opportunity permutations, recipient variates, and model seeds. The perturbation changes only the declared factor. During a partition, cross-community paths are removed and remaining row weights renormalized. During corruption, categorical packet fields from the preassigned even-index sender half are flipped after the agent constructs the packet; the agent-owned decision and outbox are not rewritten. Restoration resets fields and delivery exactly and disables the channel transform.

B Information estimators

Let Y_t denote the fixed coarse macrostate word. For observed counts $n(y_{t-k:t})$, the order- k conditional entropy estimator is

$$\hat{h}_k = - \sum_{y_{t-k:t-1}} \hat{p}(y_{t-k:t-1}) \sum_{y_t} \hat{p}_\delta(y_t | y_{t-k:t-1}) \log \hat{p}_\delta(y_t | y_{t-k:t-1}), \quad (14)$$

where δ is the frozen pseudocount. Total correlation is evaluated on the joint belief-action word over the same rolling window. These plug-in quantities describe the chosen

coarse-graining; neither is extrapolated to an unobserved microscopic entropy.

The reversal estimator forms length- L words, computes smoothed forward and reverse empirical probabilities on their union support, and divides the KL divergence by $L - 1$ attempted-update intervals. Two hundred within-trajectory time permutations establish the finite-length occupancy-matched floor. The primary adjusted statistic subtracts that mean without truncation, so negative values are possible. Confidence intervals resample complete matched clusters, not overlapping words.

C Trajectory-level representation comparison

The disruption classifier receives one feature vector for each complete panel, constructed from baseline-to-disruption changes and recovery residuals. All four panels from one graph/environment cluster are withheld together. Scaling and multinomial-logistic coefficients are fit only on the other clusters. Thus windows from a held-out trajectory never enter model fitting, nominal-manifold estimation, or feature selection. The simple, order-only, and full feature sets are frozen in the protocol; their comparison is an ablation of collective representation, not a controller competition.

D Agent autonomy and information-boundary audit

Agent autonomy is an experimental constraint rather than a label attached to the simulator. Each identity is represented by a separate state object and a separate bounded inbox, outbox, memory list, private field, workload, belief, action, confidence, and commitment. Prompt construction accepts the scheduled identity and a read-only view of that object’s delivered information. It has no argument through which evaluator-only X_t , peer memory, or an undelivered packet can enter. The scheduler supplies only an update opportunity and a recipient draw. It does not supply a proposed scientific state.

Three counterfactual tests enforce this boundary. First, perturbing one agent’s private field or memory may change that agent’s serialized prompt but must leave every peer-private digest invariant. Second, a deliberately invalid model object may trigger one syntax repair, after which failure produces a recorded no-change transition rather than a central replacement. Third, partition and restoration modify only the delivery operator: agent-owned outboxes and state remain unchanged. Formal trajectories record peer-mutation counts, message opportunities, delivered packets, complete wire bytes, and content-addressed raw-decision digests. Any privacy mutation or scheduler substitution invalidates the run rather than becoming an exclusion.

Latent alternatives are counterbalanced independently of the scientific factors. Display order, the mapping from displayed words to ± 1 , initial beliefs and actions, and private-field signs are balanced by frozen seeds. The model output is decoded only after schema validation. Consequently, a display-side preference remains measurable as a nuisance response but cannot systematically define high coupling, high noise, memory, or disruption arms.

E Quench operators and matched restoration

Let $\mathcal{P}_0$ denote the baseline transition operator induced by the fixed graph, local fields, update schedule, and clean message transform. Each quench replaces exactly one component during sweeps 16–30 and restores the original object for sweeps 31–45. Field reversal uses

$\mathbf{h} \mapsto -\mathbf{h}$ while leaving communication unchanged. The partition uses a masked delivery matrix

$$W_{ij}^{\text{part}} = W_{ij} \mathbf{1}[c(i) = c(j)] \quad (15)$$

followed by row normalization over remaining eligible neighbors. Opportunity counts remain fixed even when a cross-community delivery is unavailable. Message corruption applies a deterministic content transform to the preassigned half of senders after packet serialization and before delivery; the number and encoded length of packets are unchanged. The undisturbed arm retains $\mathcal{P}_0$ throughout.

Recovery is therefore a return of the exogenous operator, not a reset of the agent configuration. Residual displacement measures state retained by the collective after the drive is removed. Because these are abrupt, known interventions, departure at or after the boundary is a response measure and cannot be interpreted as prospective early warning.

F Frozen hypothesis tests and engineering audit

The V14 engineering pilot was limited to parser validity, latent occupancy, transition-direction counts, perturbation timing, privacy, delivery, runtime, and token projection. It comprised 256 decisions and was prohibited from inspecting quench effects. H2 uses one maximum-departure contrast per matched cluster; H3 uses one within-field-reversal early-minus-late restoration contrast per cluster; H4 uses one full-minus-order-only held-out accuracy contrast per cluster. Exhaustive sign flips enumerate all 2^6 paired null assignments. Ten thousand cluster-bootstrap draws provide intervals, and Holm correction controls the H2–H4 family.

Two retained pilot process failures occurred before protocol freeze. The first stopped before any decision because deterministic CUDA execution lacked the documented cuBLAS workspace variable. The second completed all 256 decisions but encountered a configuration-key mismatch during summary construction. Rather than exceed the 300-decision pilot ceiling, the retained responses were content-addressed and deterministically replayed, validating every prompt, seed, and sampling temperature. No prompt, scientific threshold, perturbation, estimator, or formal seed changed. The frozen protocol records both the pilot summary and the execution-source checksum.

An outcome-blind source audit after the freeze identified two limitations that were not repaired post hoc. Although three- and seven-sweep macrostate series are saved, nominal-distance sensitivity was implemented only for the primary five-sweep series. Observable-family deletion was implemented, but deletion of each single observable was not. These omissions do not change the frozen H2 or H4 primary calculations; the absent sensitivities are reported as missing rather than reconstructed after observing formal outcomes.

G Nominal manifold and trajectory geometry

For held-out quench cluster g , the nominal center $\hat{\mu}_{-g}$ and ridge-regularized covariance $\hat{\Sigma}_{-g}$ are estimated only from undisturbed trajectories in the other clusters. The standardized departure is

$$d_{g,t} = \left[(Z_{g,t} - \hat{\mu}_{-g})^T \hat{\Sigma}_{-g}^{-1} (Z_{g,t} - \hat{\mu}_{-g}) \right]^{1/2}. \quad (16)$$

Its nominal 95th percentile is likewise learned without cluster g . Maximum departure, time outside that region, final residual, and the first two consecutive recovery points

below threshold are computed at trajectory level. The energy–entropy loop area uses the polygonal path in temporal order and is a finite dynamical hysteresis descriptor; it is not thermodynamic work.

The representation comparison uses prospectively fixed trajectory summaries. All four conditions sharing one graph and environment are withheld together. A fixed L_2 -regularized multinomial logistic model and training-fold standardization prevent the full representation from receiving a different model family. The analysis reports balanced accuracy, macro recall, confusion matrices, graph-level predictions, feature-group deletions, and the paired full-minus-order-only contrast. With six held-out clusters, coefficient patterns remain finite-sample and are not promoted to universal biomarkers.

H Estimator and coarse-graining sensitivity

The primary block length is three and pseudocount is 0.5. Sensitivities use block lengths two and four and pseudocounts 0.1 and 1.0. Configuration bins are fixed before formal execution. Time-shuffled paths preserve length and marginal occupancy while destroying temporal order; their mean reversal divergence is subtracted as the primary bias floor. These corrections can yield a slightly negative adjusted value, which is retained rather than truncated.

I Additional candidate figures

The paper package contains 26 candidate vector figures with exact source CSVs. Supplemental candidates include micro-to-macro mappings, condition-specific phase portraits, network snapshots, memory block sensitivities, bias floors, conditional-memory depth, recovery summaries, cluster confusion matrices, topology context, and a claims forest. The figure catalog records each plot’s scientific purpose, source table, suggested role, supported claim, and limitation. Selection for submission should follow scientific review rather than visual favorability.

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